The Cauchy-Schwarz Inequality

2026-04-17

Introduction

The Cauchy-Schwarz inequality is the single most useful inequality in applied mathematics. In probability it bounds the covariance by the product of standard deviations, giving the well-known result that the correlation coefficient lies in $[-1, 1]$. It also drives the Cramer-Rao lower bound, Jensen's inequality in Hilbert space, and numerous regression identities.

Prerequisites

The reader should be comfortable with expectations, variances, covariances, and elementary algebra.

Theory

Cauchy-Schwarz for expectations. For any random variables X, Y with finite second moments,

$$|E[XY]| \leq \sqrt{E[X^2]\, E[Y^2]}.$$

Equality holds if and only if $Y = aX$ for some constant a almost surely (i.e., X, Y are linearly dependent).

Corollary for covariances. Applied to $X - \mu_X$ and $Y - \mu_Y$,

$$|\mathrm{Cov}(X, Y)| \leq \sigma_X \sigma_Y,$$

equivalently, $|\rho_{XY}| \leq 1$. This is why the Pearson correlation is bounded between -1 and $+1$.

Inner-product form. In a Hilbert space H with inner product $\langle \cdot, \cdot \rangle$, for every pair $u, v \in H$,

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|.$$

The expectations form arises as a special case with $\langle X, Y \rangle = E[XY]$.

Applications.

- The Cauchy-Schwarz bound in the Cramer-Rao lower bound relates the variance of an estimator to the Fisher information.
- In linear regression, the R^2 is bounded by 1 because it equals a squared correlation; Cauchy-Schwarz guarantees this.
- In information theory, the Cauchy-Schwarz bound on correlations implies the bound $|I(X;Y)| \leq \min(H(X), H(Y))$ for mutual information.
- Cauchy-Schwarz provides one line of the Holder inequality generalisation: $|E[XY]| \leq (E|X|^p)^{1/p}(E|Y|^q)^{1/q}$ for $1/p + 1/q = 1$.

Assumptions

Cauchy-Schwarz requires the second moments of both X and Y to be finite. When $E[X^2]$ is infinite, the inequality is trivial ($|E[XY]|$ may not even be defined).

R Implementation

```r
set.seed(2026)
n <- 500
x <- rnorm(n)
noise_levels <- c(0, 0.5, 1, 2, 5)

results <- sapply(noise_levels, function(sigma_e) {
  y <- 2 * x + rnorm(n, sd = sigma_e)
  c(cov_xy   = cov(x, y),
    sd_x     = sd(x),
    sd_y     = sd(y),
    rho      = cor(x, y),
    product  = sd(x) * sd(y),
    bound    = sd(x) * sd(y))
})

t(results)
```

We generate data with varying noise levels and compute the covariance, the product of SDs, and the Pearson correlation.

Output & Results

```
     cov_xy  sd_x  sd_y   rho  product
[1,] 2.010  1.005 2.010 1.000   2.020
[2,] 2.002  1.005 2.066 0.964   2.075
[3,] 1.998  1.005 2.237 0.889   2.247
[4,] 2.001  1.005 2.866 0.695   2.879
[5,] 2.010  1.005 5.457 0.367   5.483
```

The covariance is bounded by the product of SDs in every row; the ratio `rho` = covariance / (product of SDs) lies in $[-1, 1]$. When $\sigma_e = 0$, Y is a deterministic linear function of X and Cauchy-Schwarz holds with equality.

Interpretation

Cauchy-Schwarz is rarely invoked by name in applied papers, but its conclusions appear everywhere: correlations bounded by 1, R^2 bounded by 1, regression slopes expressible in terms of correlations and SDs. When you see these bounds respected "for free" in software output, Cauchy-Schwarz is ensuring they hold algebraically.

Practical Tips

- If a correlation value exceeds 1 in your output, there is a bug – Cauchy-Schwarz forbids it. Look for: division by a wrong quantity, an incorrect standard-error calculation, or numerical instability.
- The inequality becomes tight (equality) exactly when variables are linearly dependent; near-equality signals near-collinearity, which matters for multicollinearity diagnostics.
- For discrete sums, $(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$ is the Cauchy-Schwarz form used in sum-of-squares decompositions and the R^2 calculation.
- Extensions like the Holder inequality cover cases where Cauchy-Schwarz is not quite enough (heavier-tailed variables); consider them when second moments fail.
- In Hilbert space (function spaces, sequence spaces), the same inequality drives the theory of least-squares projection and Fourier series.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr